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Mitigating TGO growth with glass-ceramic based thermal barrier coatings for gas turbine applications



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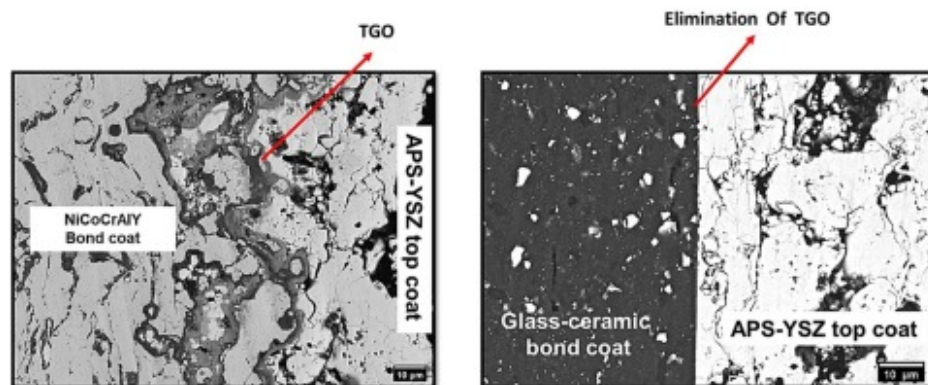


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Abstract

Thermally grown oxide (TGO) layer develops and grows at metal based bond coat (BC) and ceramic top coat (TC) interface of thermal barrier coating (TBC) system at elevated temperatures and mainly composed of aluminium oxides (Al_2O_3). In addition to providing efficient protection from oxidation and hot corrosion, TGO layer increases effective adhesion between BC and TC. Failure in TBC systems occurs by ceramic top coat spallation due to crack formation at top coat-TGO interface or bond coat-TGO interface due to abnormal progression of TGO layer. However, research is essential to derive an in-depth understanding of TBC system failure. This work investigated the oxidation behaviour of a typical TBC system fabricated by replacing conventional (MCrAlY) bond coat with glass-ceramic coating. Three TBC systems were developed: glass-ceramic bond coated TBC system, composite (15 wt% YSZ reinforced glass-ceramic) bond coated TBC system, and double layered bond coated (glass-ceramic coat and composite coat) TBC system with 8-YSZ top coat. Oxidation resistance was assessed through isothermal oxidation tests at 1000°C for 100 h. Results indicated that all glass-ceramic based TBC systems exhibited promising oxidation resistance while triple layered TBC system showed the best performance. Microstructural study showed the absence of detrimental TGO layer that made these coatings advantageous for high temperature applications.

Graphical Abstract



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Keywords

Glass-ceramics; ZrO₂; Microstructure; Thermal properties; Engine components

1. Introduction

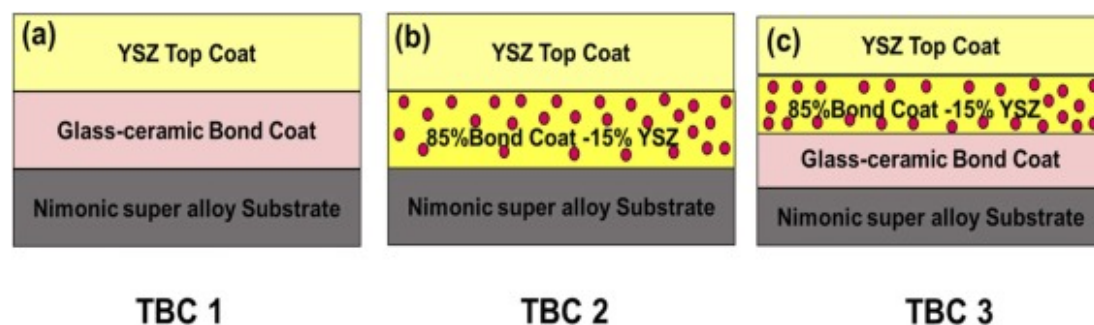
Thermal barrier coatings (TBCs) indeed enhance performance and durability of components of turbine engines used in aerospace applications. Engine components are shielded from intense and prolonged heat loads by these 100 µm-2 mm thermal insulation layer [1], [2]. A typical TBC system structure is composed of metal substrate, metal bond coat and ceramic top coat. Super alloy substrate forms the base of the component and provides the structural integrity necessary for the turbine part. Super alloys are specifically designed to withstand high temperatures and mechanical stresses, making them ideal for turbine applications where both factors are critical. Super alloy substrates are preferred for their ability to maintain high yield strength even at temperatures exceeding 850 °C [3], [4], [5]. Metal bond coat (BC) enhances adherence of ceramic top coat to substrate and acts as a diffusion barrier to prevent interdiffusion between substrate and ceramic layer [6]. Most commonly preferred bond coat material is an alloy of metals designated as MCrAlY [7]. The top coat (TC) is outermost layer is made of ceramic materials such as yttria stabilized zirconia (YSZ). Ceramic top coat affords excellent thermal protection, reducing surface temperature of the component and thus improving its thermal efficiency and durability [8], [9], [10]. Each of these layers in a TBC system is carefully engineered to provide specific properties that collectively enhance the performance and longevity of turbine components. During operation, MCrAlY bond coat material undergoes oxidation leading to development of thermally grown oxide (TGO) layer between bond coat and top coat [11]. TGO consists of oxides and grows as bond coat oxidizes at high temperatures. TGO serves as stress relieving layer that helps to protect bond coat and substrate from stresses induced by temperature gradients [12], [13].

Thermal barrier coatings face a range of challenges including mechanical stress, thermal shock, and sintering. These coatings consist of layers with different properties leading to the development of residual stresses. These stresses can cause issues such as the separation of layers between top coat and TGO, or between TGO and bond coat. On the other hand, continuous service and

oxidation can lead to progressive degradation of bond coat material eventually hindering its ability to form the protective α - Al_2O_3 rich TGO layer. This degradation results in formation of oxides like Cr_2O_3 and NiO and ultimately leads to failure [14], [15], [16]. Though structural integrity of TBC system determines its operational lifetime still TGO layer significantly contributes to system failure. Thickness of TGO layer is crucial in predicting the remaining lifespan of a TBC system influenced by repeated thermal cycling during service [17], [18], [19]. One potential strategy to enhance the operational lifespan of TBC systems involves controlling development of TGO or promoting preferential formation of α - Al_2O_3 [20]. Many researchers have attempted to limit the TGO layer thickness for an ideal TBC system [21]. Previous studies have primarily focused on investigating the processes involved in forming TGO through heat treatment of MCrAlY-coated samples [22], [23], [24]. Takahashi et al. [25] observed that growth of TGO layer is influenced by heat treatment in both air and vacuum environments. Further, they found out that surface roughness has an important role in top coat adhesion. Das et al. [26] explored use of glass-ceramic bond coat to address issues related to uncontrolled growth of TGO layer. Glass-ceramics is one kind of material formed through crystallization of glass resulting in crystal formation within an amorphous glassy matrix. Glass-ceramics typically exhibit attractive properties such as low porosity, high strength, toughness, high temperature stability and high chemical durability [27], [28]. These materials can be tailored to achieve required properties [29], [30], [31]. Das et al. [32] specifically studied thermal cycling behaviour of one TBC system having glass-ceramic bond coat. Their findings highlighted that this system exhibited superb thermal shock resistance and ensured strong adhesion of coating to substrate. Additionally, hot corrosion behaviour of glass coating on metal substrate was also studied at 1000°C . These coatings successfully withstood the corrosive environment up to 35 h wherein the coating/substrate interface region was marginally affected and mass loss was found to be negligible [33]. Pallabi et al. [34] observed better performance of glass-ceramic bond coated TBC system over traditional TBC system during long thermal cycling test at 1000°C for 500 cycles. No TGO layer was observed in new TBC system [34]. Functionally graded three layered glass-ceramic based TBC system exhibited good oxidation resistance up to 100 h at 1100°C . It was also noted that these coatings had good adherence to the substrate [35].

Current work primarily proposes a strategy to eliminate TGO layer formation in TBC system by replacing MCrAlY coatings with glass-ceramic based coatings. For this study, three different TBC systems were fabricated in which TBC1 and TBC2 are a duplex layered systems consisting of glass-ceramic bond coat and 15 wt% YSZ-85 wt% glass-ceramic composite bond coat, respectively along with YSZ (yttria stabilized ZrO_2) ceramic top coat. TBC3 is a double bond coated TBC system fabricated by stacking above mentioned composite bond coat over glass-ceramic bond coat followed by YSZ top coat. Schematic representation of three TBC

systems is shown in Fig. 1. It is expected that by creating novel TBC3 system, functional properties would tend to gradually change from bond coat to top coat and thereby, stabilizing entire TBC system.



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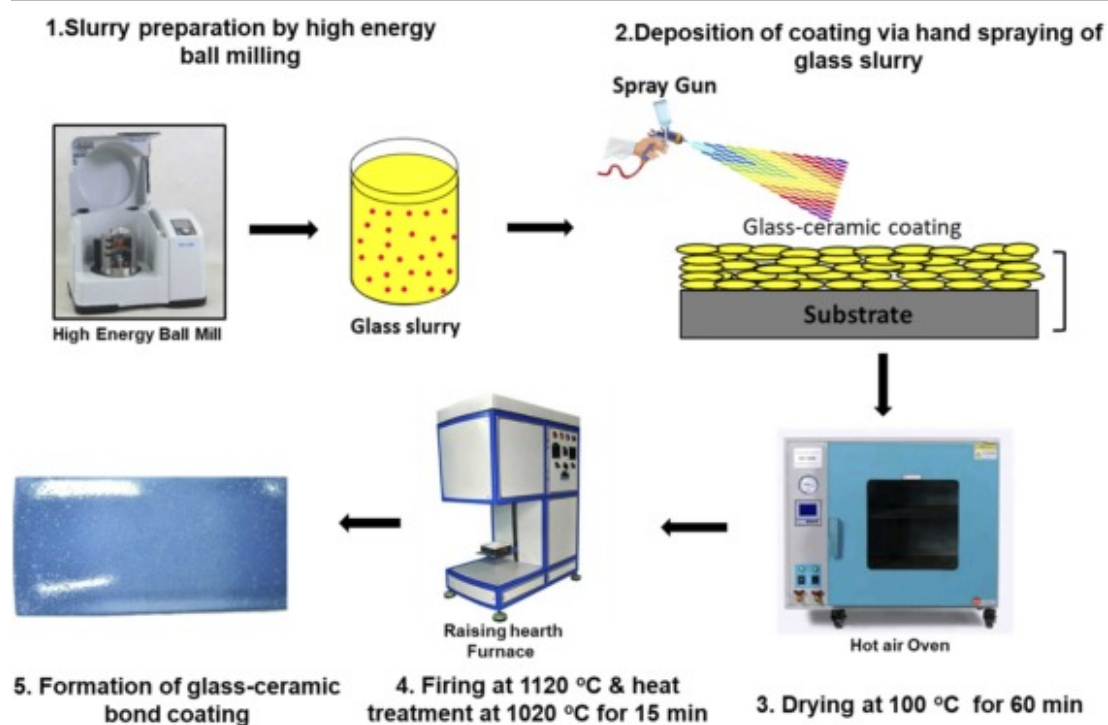
Fig. 1. Schematics of developed TBC systems.

2. Materials and methodology

2.1. Preparation of TBC systems

Commercially available nimonic super alloy (CTE value- $13.85 \times 10^{-6}/^{\circ}\text{C}$ from room temperature to 600°C) and plasma sprayable 8 wt% yttria doped ZrO_2 (YSZ) powder (CTE value- $7.9 \times 10^{-6}/^{\circ}\text{C}$) were used as substrate and top coat material for the TBC system. To prepare the substrate surface for coating, degreasing was done at 900°C for 1 h to remove surface contaminants followed by grit blasting with alumina beads to enhance mechanical interlocking for better coating adhesion. Subsequently, the grit-blasted substrates were cleaned ultrasonically with acetone to eliminate any remaining impurities. The precursor for coating material consisted of glass-ceramic frits with the following nominal composition (wt%): MgO (35–42 %), SiO_2 (40–46 %), Al_2O_3 (15–22 %), and TiO_2 (10–14 %). For the preparation of composite powder, the glass-ceramic frits were mixed with 8-YSZ (8 wt% yttria stabilized zirconia) powder in a ratio of 8.5:1.5 and ball milled for 72 h. The thermal properties of developed TBC system were studied in detail. Glass-ceramic coating and composite coating (85 wt% glass-ceramic–15 wt% YSZ) used in this study had thermal expansion coefficient value of about $7.5 \times 10^{-6}/^{\circ}\text{C}$ and $8.9 \times 10^{-6}/^{\circ}\text{C}$, respectively. Both the composite coating and glass-ceramic coating were prepared using enamelling technique. This method involves four main stages:

- (i) Slurry Preparation: The coating slurry was prepared by wet milling the glass-ceramic/composite powder with 5 wt% of china clay and water.
- (ii) Coating application: Glass slurry was applied on grit-blasted substrates by hand spray gun.
- (iii) Drying: Subsequently, the slurry-coated substrates were dried at 100°C for 2 h before firing to remove moisture from green coating.
- (iv) Firing: The coated samples were fired at a temperature range of 1120°C-1160°C for 15 min and thereafter additional heating was done at 1050°C for 1 h. A dense and impermeable glass-ceramic or composite coating was formed on metallic substrate. The pictorial representation of the enamelling technique is given in Fig. 2.



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Fig. 2. Pictorial representation of enamelling technique.

Finally, 8-YSZ top coat was applied by using atmospheric plasma spraying technique on glass-ceramic or composite coated substrates. Plasma spray conditions of YSZ top coat have been mentioned in [Table 1](#).

Table 1. Plasma spraying parameters for APS-YSZ top coat.

Powder injection outside nozzle	6 mm
Power input (kW)	50
Primary/secondary gas in the plasma (standard l/min)	35 Ar/10H ₂
Carrier gas (Ar) flow in the feeder Nozzle (standard l/min)	2.6
Stand-off distance (mm)	150

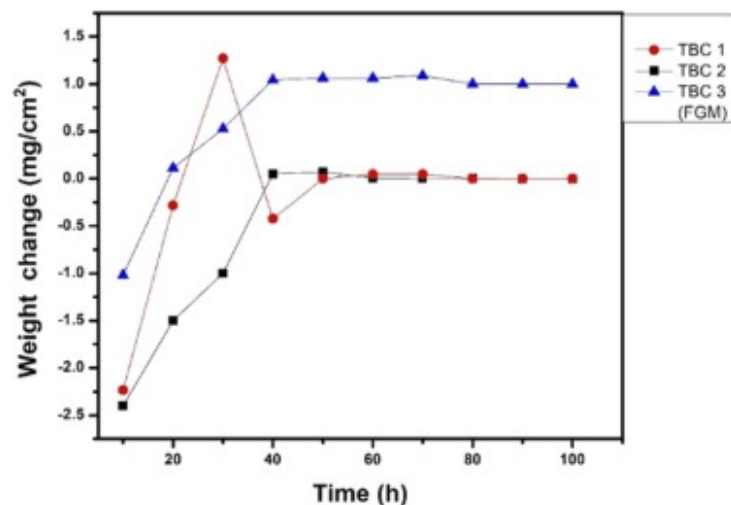
2.2. Isothermal oxidation test

During oxidation test, two specimens from each TBC systems were heated at 1000 °C for 100 h in air. All specimens were loaded in the muffle furnace. During initial 10 h, the specimens were cooled to room temperature for 20 min to evaluate oxidation after one hour. Weight gain of specimens was measured by analytical balance (accuracy-0.1 mg). Procedure was followed until cumulative 100 h was achieved. Cross-sectional microstructures of specimens were studied using Field Emission Scanning Electron microscopy (Supra 35VP, Carl Zeiss, Germany) followed by elemental composition analysis via EDX analysis. The phase changes occurring in each layer of TBC system were evaluated by X-ray Diffraction analysis (Rigaku, Ultima IV). For FESEM observations, samples were cut by diamond cutter followed by mounting. The mounted specimens were sliced and ground by abrasive papers of 1200 grit. After that, sample polishing was conducted using diamond paste of 3 micron followed by 1 micron for 15 min each. Mechanical properties of the coatings were measured by a nano-hardness tester (Hysitron TI 950, Triboindenter, USA) using 9000 µN load. Loading and unloading rate was maintained to 0.5 µN/Sec.

3. Results and discussion

3.1. Oxidation kinetics and its implication

The nimonic alloy substrate was oxidized and formed a very thin oxide scale at the interface of substrate and glass coating during heat treatment of glass powder coated substrate. Proper adhesion of glass coating to substrate was achieved due to formation of oxide layer at interface. The thin oxide layer formed between substrate and bond coat protected the substrate from further oxidation also [26], [36]. The interfacial adhesion of TBCs is one of the key factors affecting their stability and durability in high-temperature environments. Researchers have evaluated interfacial adhesion of coating with the substrate by various methods [37], [38], [39], [40]. In our case we utilized 90° bend test as qualitative method for evaluating interfacial adhesion of as-fabricated coating with the substrate. Coating did not peel off from the substrate after the test. During isothermal oxidation of the TBC system the oxide layer present at the interfacial region of substrate and coating got extended slightly due to diffusion of aluminium, nickel, chromium, silicon from the substrate towards substrate-bond coat interface and reaction with oxygen penetrated through top coat and bond coat from atmosphere or oxygen ion of glass-ceramic based bond coat resulting in thickening of oxide scale to certain extent. This oxide reaction layer thickening led to weight gain of the TBC system [33], [34], [35], [36]. Fig. 3 shows oxidation kinetic curves of three TBC specimens after heating isothermally at 1000 °C for 100 h. It was noticed that weight of TBC 1 system was increased in the first 30 h and thereafter there was a sudden drop in weight followed by nearly constant weight after 50 h of oxidation as shown in Fig. 3. The kinetic curve of TBC 2 system exhibited a gradual increase of weight until 40 h, after which the weight change line graph resembled almost no weight change with respect to time (Fig. 3). Functionally graded multilayered TBC 3 system demonstrated similar weight change trend like TBC 2 (Fig. 3).



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Fig. 3. Isothermal oxidation kinetic curves of three different TBC specimens.

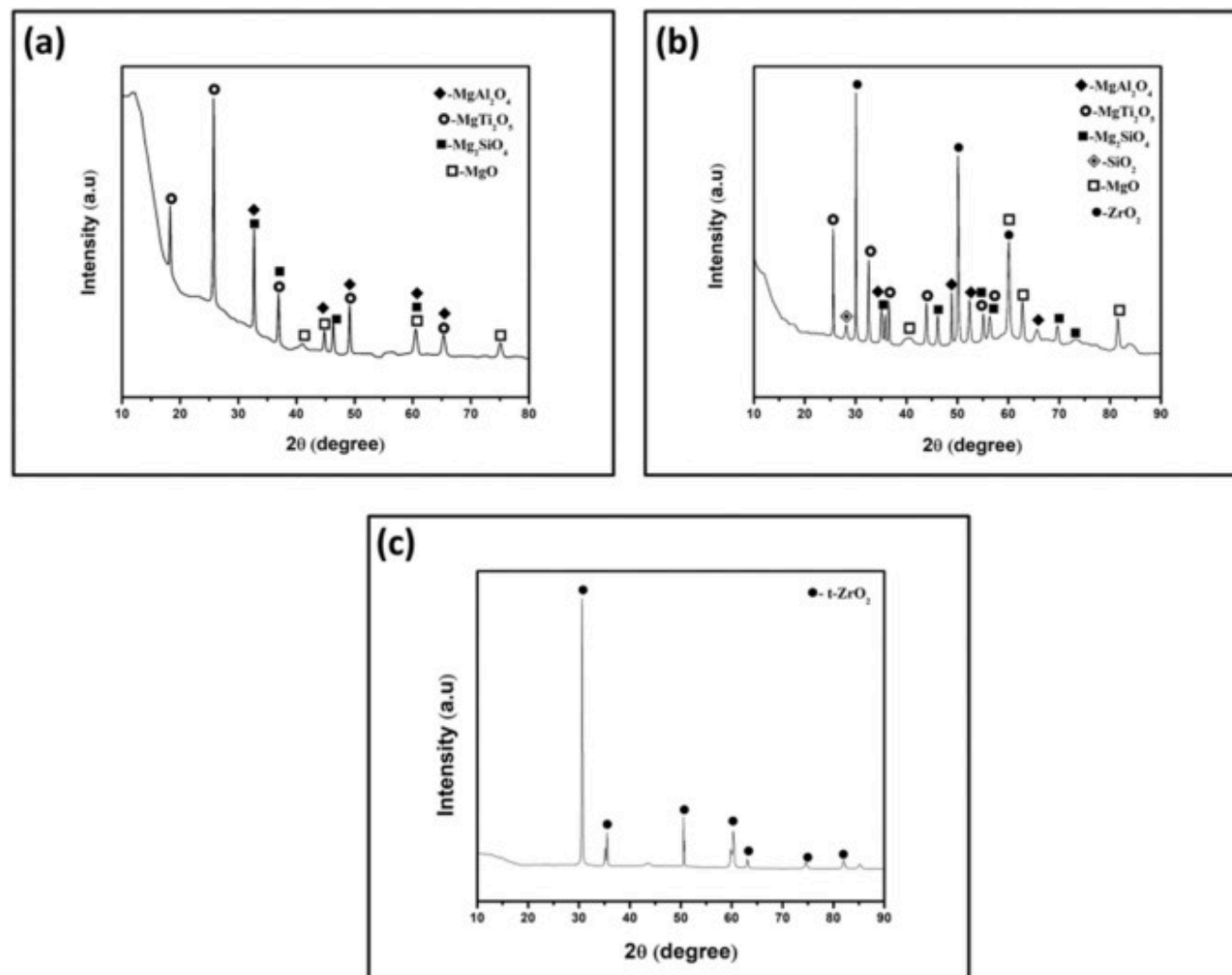
It can be found that the weight gain occurred during oxidation at 1000 °C in all the three different TBC systems due to formation of oxide scale between substrate and bond coat as was observed in our previous study [33]. However, sudden drop in weight after 30 h in pure glass-ceramic bond coated TBC1 system might be attributed to the negligible chipping of top coat, which dominated over oxide layer formation at the substrate-bond coat interface. It was noteworthy that weight change behaviour of TBC2 and TBC3 systems was identical, showing initial increase in weight as oxidation influenced more than top coat disintegration. Later, weight change was almost constant with respect to time after 40 h of oxidation on account of relatively balance between top coat chipping and oxide layer formation. On the contrary, no abrupt drop in weight was observed in these TBC2 and TBC3 systems. However, weight gain per unit surface area of TBC3 system was relatively higher than that of TBC2 system. This indicated lesser tendency to top coat disintegration in case of TBC3 system. It is evident from oxidation kinetic study that triple layered TBC3 system is most dependable among three TBC systems. Abrupt change in weight indicated inferiority of TBC1 system with glass-ceramic bond coat compared to other TBC systems.

In case of TBC2 and TBC3 systems thermal expansion mismatch effect was smaller than that for TBC1 system leading to reduced tendency to top coat chipping. The different thermal expansion coefficient (CTE) values for dissimilar bond coats of the TBC

systems were responsible for the discrepancy noted in weight change behaviour. The CTE values of glass-ceramic bond coat and composite bond coat were about $7.5 \times 10^{-6}/^{\circ}\text{C}$ and $8.9 \times 10^{-6}/^{\circ}\text{C}$ whereas these values were about $13.85 \times 10^{-6}/^{\circ}\text{C}$ and $7.9 \times 10^{-6}/^{\circ}\text{C}$ for nimonic alloy substrate and YSZ top coat. Thus, TBC3 system was most reliable among three TBC systems in terms of weight change behaviour during isothermal oxidation.

3.2. Phase analysis

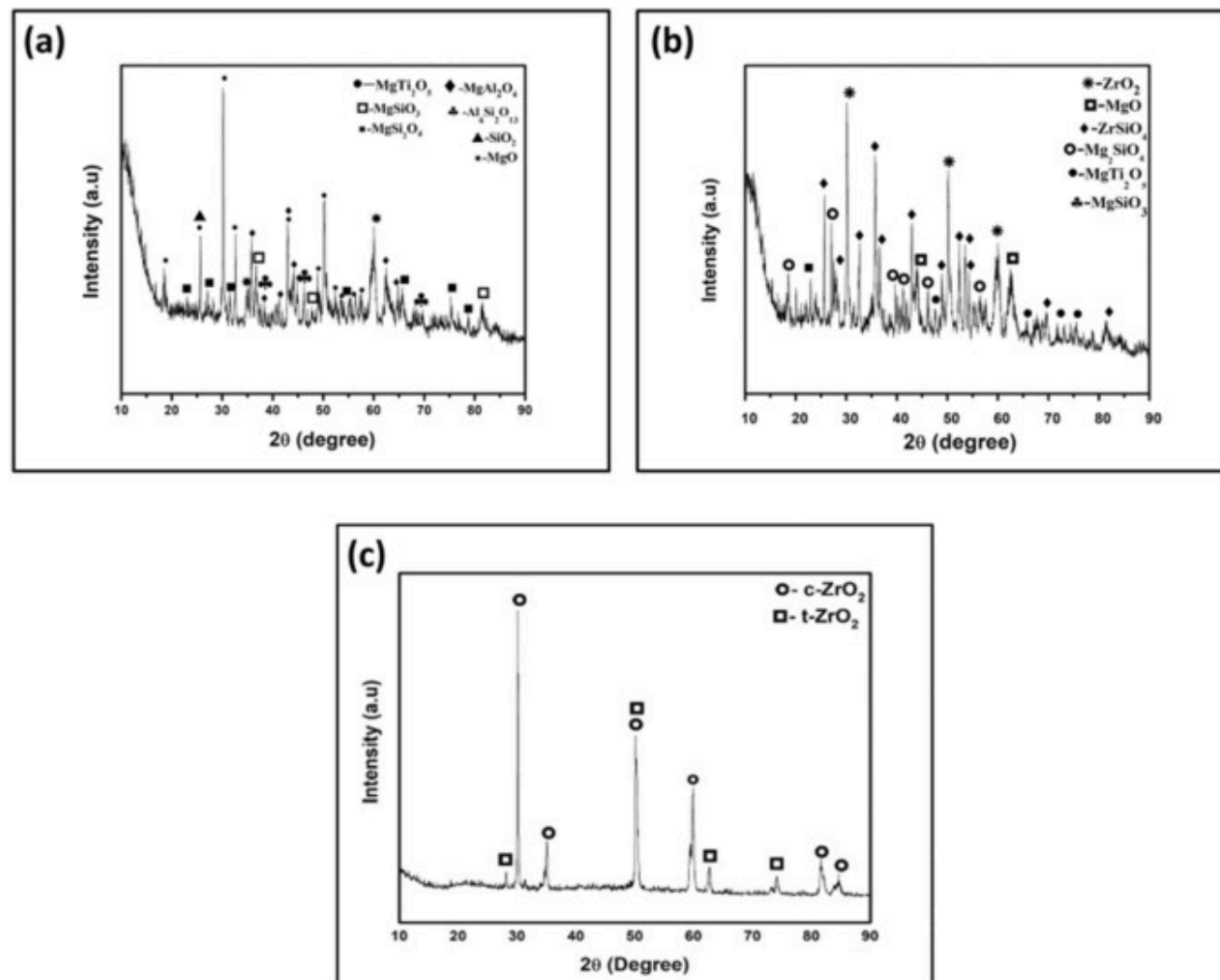
As prepared glass-ceramic coating showed presence of periclase, magnesium aluminate, magnesium silicate and magnesium titanate phases. On the other hand, as prepared composite coating had periclase, magnesium aluminate, magnesium silicate, magnesium titanate and zirconia phases whereas tetragonal zirconia was detected on top coat ([Fig. 4a-c](#)). [Fig. 5a](#) illustrates XRD patterns of glass-ceramic bond coat following oxidation at 1000°C for 100 h. Main diffraction peaks present in the pattern were pseudobrookite (MgTi_2O_5), magnesium silicate spinel (MgSi_2O_4) and magnesium aluminate spinel (MgAl_2O_4) along with traces of mullite phase ($\text{Al}_6\text{Si}_2\text{O}_{13}$) and magnesium silicate phase (MgSiO_3). Additionally, there was significant presence of periclase phase (MgO) and a small amount of silica (SiO_2). The formation of mullite at lower temperatures than typically formation temperature (1600°C) was attributed to prolonged exposure at 1000°C [\[41\]](#). Additionally, cristobalite- SiO_2 was obtained in lieu of low quartz- SiO_2 , indicating no detrimental phase transformation occurred even after prolonged oxidation at 1000°C . [Fig. 5b](#) depicts the XRD pattern of YSZ– glass-ceramic composite coat after oxidation at 1000°C for 100 h. After isothermal oxidation, composite coating exhibited similar phases as the glass-ceramic bond coat including magnesium titanate, periclase, and magnesium silicate spinel (Mg_2SiO_4) with an inverse spinel crystal structure. Interestingly, mullite phase was entirely absent and weak peaks corresponding to the magnesium aluminate spinel phase were observed. Additionally, prominent peaks of zircon phase confirmed the occurrence of zircon formation reaction between YSZ inclusions and Si-O network in glass matrix due to prolonged oxidation at high temperatures. [\[42\]](#). Furthermore, peaks of tetragonal zirconia affirmed the presence of residual ZrO_2 particles that had not reacted with glassy matrix. Precisely, zircon peaks were visible instead of YSZ peaks with increasing oxidation time due to interfacial reactions between YSZ and glass matrix.



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Fig. 4. XRD diffraction patterns of as fabricated (a) glass-ceramic bond coat, (b) glass-ceramic/YSZ composite coat and (c) 8-YSZ top coat.



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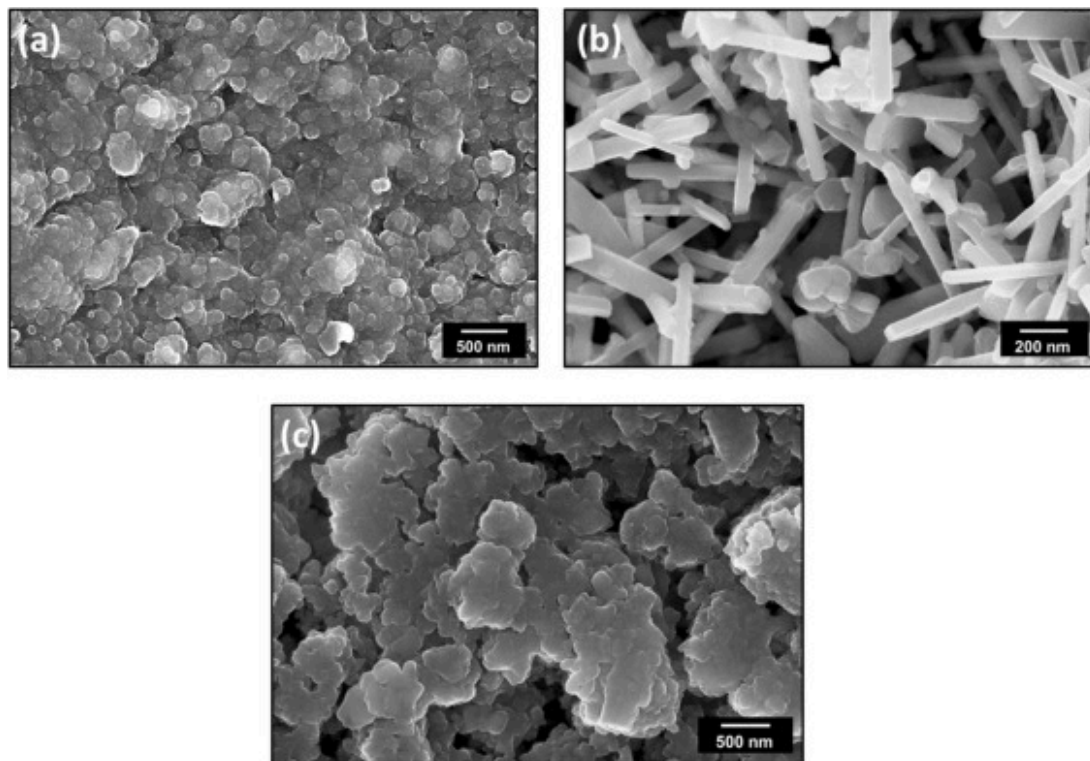
Fig. 5. XRD diffraction patterns of (a) glass-ceramic bond coat, (b) glass-ceramic/YSZ composite coat and (b) 8-YSZ top coat after isothermal oxidation at 1000 °C for 100 h.

From the XRD investigation (Fig. 5c) it was observed that tetragonal crystal structure of zirconia (t- ZrO_2) were present along with emergence of few peaks related to cubic crystal structure of zirconia (c- ZrO_2). This indicates that structural changes occurred in

the top coat due to prolonged soaking time at 1000 °C. However, monoclinic phase transformation of zirconia has been reported as detrimental due to volume changes of up to 4 % that induces stress in the TBC system. On the other hand, cubic transformation has not yet been reported to have a negative effect on the TBC system [43]. Cubic zirconia is more suitable as TBC as it provides heat resistance, thermal expansion match with metal substrate and strain tolerance [44]. Therefore, formation of cubic zirconia enhanced the performance of TBC system during oxidation.

3.3. Morphology of coating surfaces

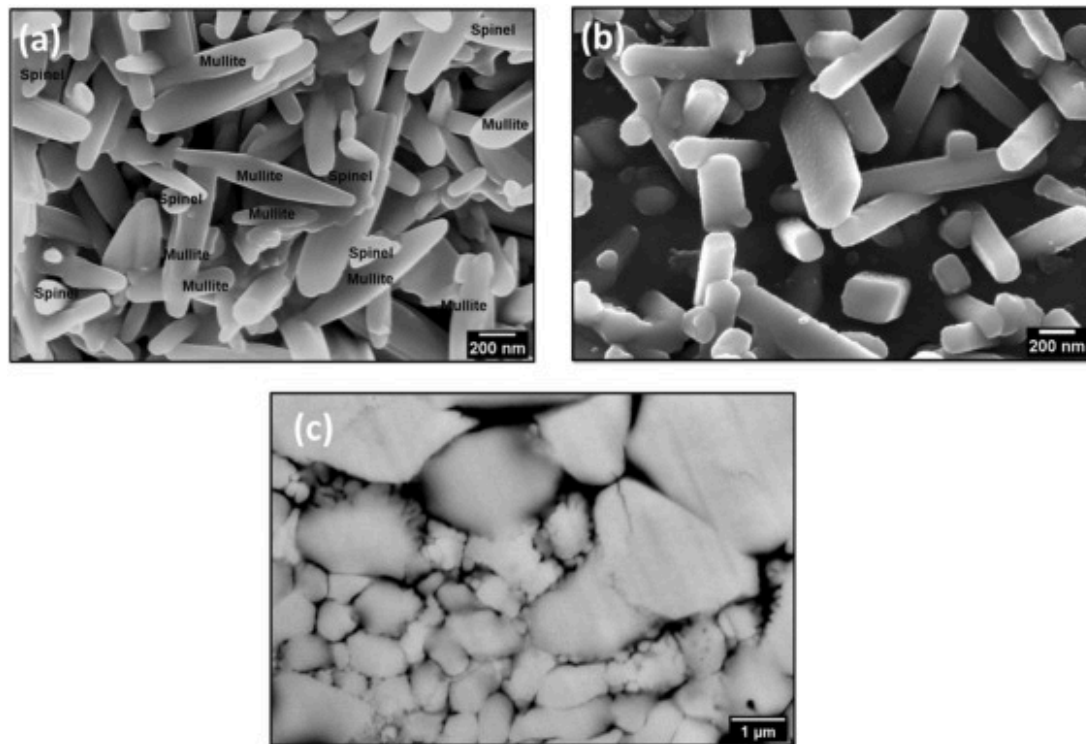
Surface morphology of as fabricated glass-ceramic bond coat, glass ceramic-YSZ composite coat and YSZ top coat has been shown in Fig. 6. Fig. 7a-c represents surface morphology of glass-ceramic bond coat, glass ceramic-YSZ composite coat and YSZ top coat after oxidation for 100 h at 1000 °C. As fabricated similar glass-ceramic coating showed presence of magnesium aluminium titanate, aluminium titanate and magnesium silicate phases in the glassy matrix [26]. From Fig. 7a acicular shaped mullite crystals can be seen in glassy matrix. Mullite (aluminium silicate) has high oxidation resistance as well as excellent thermal stability, which improved thermal performance of coating at elevated temperature [45]. Some agglomerated and spherical phase due to MgAlO_4 spinel formation was also observed in the glass matrix (Fig. 7a) as indicated by XRD data. Magnesium aluminate is resistant to corrosive agents with high temperature strength and low thermal conductivity [46]. EDX analysis results also confirmed the presence of Mg, Al, O elements. Formation of mullite and magnesium aluminate enhanced thermal barrier performance of the present TBC systems at 1000 °C. Fig. 7b demonstrates that glass ceramic-YSZ composite coating, subjected to similar conditions, developed flattened rod-shaped crystals surrounded by an amorphous glassy phase. EDX analysis revealed a matrix rich in silica, aluminium, and oxygen along with Zr elements indicating presence of YSZ particles within the composite. YSZ particle inclusion in glass matrix facilitated zircon crystal formation by creating cation vacancies. Zircon is a well-known material with high heat resistance property and chemical stability [47]. Therefore, its formation in the composite coating favoured the role of coating as heat resistant coating at high temperatures. Furthermore, residual ZrO_2 peaks as detected by XRD analysis offered the advantageous properties. Fig. 7c displays FESEM image of the well-polished surface of top coat revealing irregularly shaped coarse grains as well as finer ones. Approximately 12 % porosity was calculated using ImageJ software attributed to the sintering effect during prolonged soaking.



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Fig. 6. FSEM images of (a) glass-ceramic bond coat, (b) glass-ceramic /YSZ composite coat and (c) 8-YSZ top coat before isothermal oxidation.



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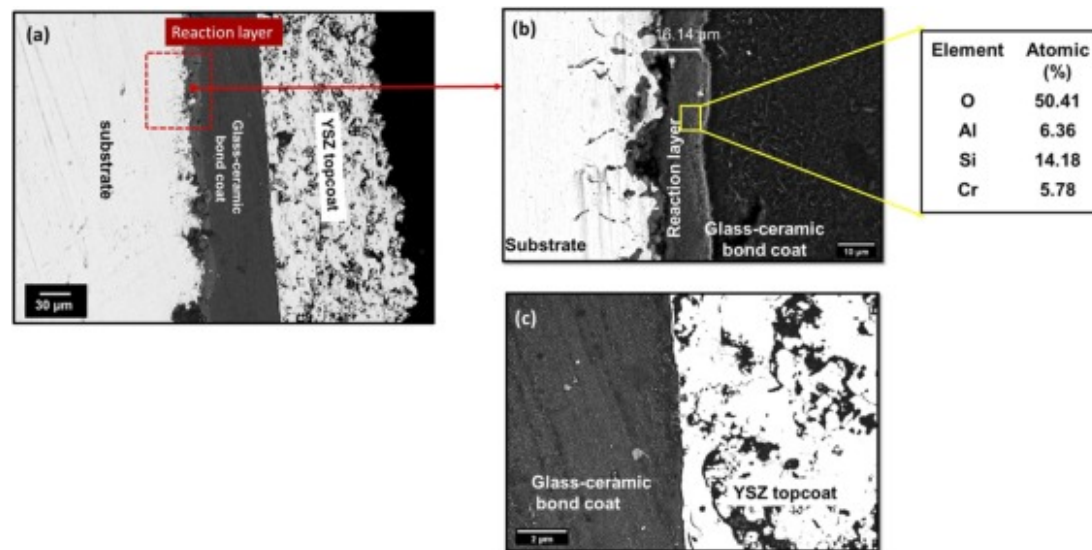
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Fig. 7. FSEM images of (a) glass-ceramic bond coat, (b) glass-ceramic /YSZ composite coat and (c) 8-YSZ top coat after isothermal oxidation at 1000 °C for 100 h.

Further, it may be noticed that crystal size of composite bond coat was larger compared to that of glass-ceramic bond coat (Fig. 7). Therefore, effective surface area of the reinforcement was decreased in case of the composite bond coat, which led to decrease in porosity. Reduced porosity resulted in lesser ingress of oxygen ion through the composite bond coat than glass-ceramic bond coat leading to lower oxidation.

3.4. Cross-sectional microstructure analysis

Fig. 8a depicts cross-section microstructure of duplex layered TBC1 system after oxidation for 100 h at 1000 °C. Two distinct coatings were clearly visible: the first, adjacent to the substrate, was glass-ceramic bond coat with a thickness of approximately 100 μm followed by YSZ top coat ($\approx 200 \mu\text{m}$) exhibiting porous lamellar structure typical of atmospheric plasma sprayed coating. Notably, a new layer was formed between substrate and bond coat, as was observed in magnified microstructure of Fig. 8a (Fig. 8c). Thickness of that layer was measured using image analysis software at multiple points and averaged to $\approx 16 \mu\text{m}$. EDX analysis was performed to investigate the composition of the reaction layer, which identified the presence of elements such as Al, Si, O, and Cr, as indicated in Fig. 8b.



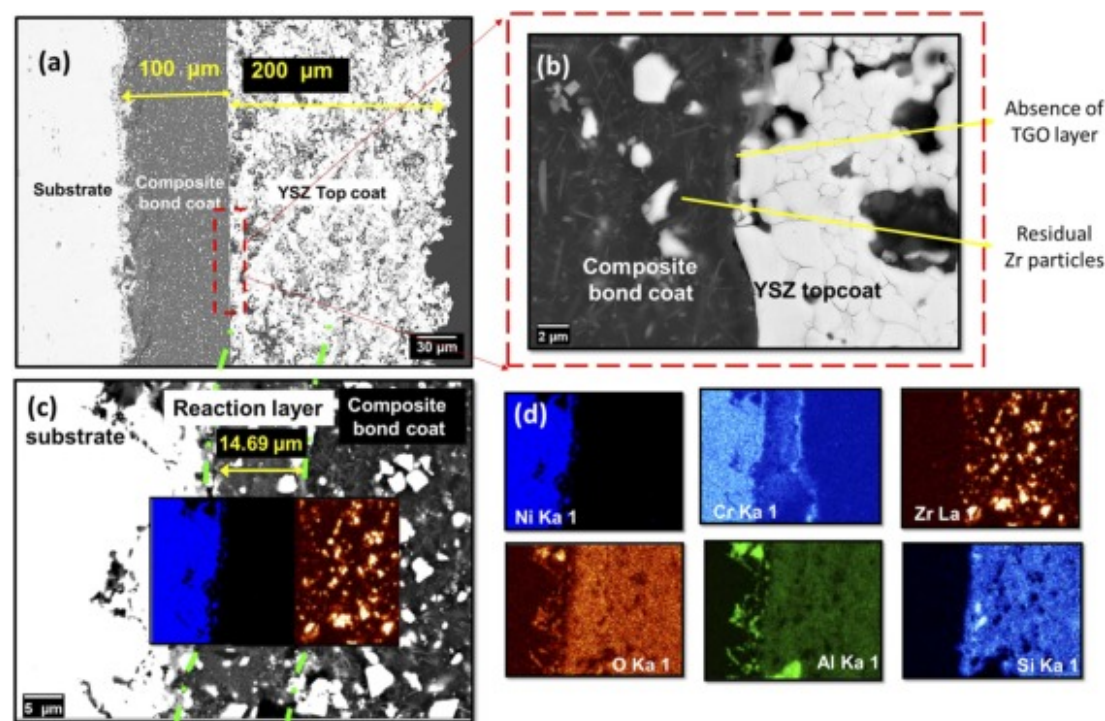
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Fig. 8. (a) Cross-sectional FESEM micrograph of glass-ceramic bond coated duplex layered TBC system, (b) substrate-bond coat interface (magnified) and (c) bond coat-top coat interface (magnified) after oxidation for 100 h at 1000 °C.

Fig. 9a presents cross-section micrograph of TBC2 system, featuring a composite bond coat comprising of 85 wt% glass-ceramic and 15 wt% YSZ. There was clear nonappearance of TGO between top coat and bond coat (Fig. 9b). However, a thin layer with an thickness of $\approx 14 \mu\text{m}$ was evident between substrate and composite bond coat, as depicted in Fig. 9c. BSE elemental mapping was conducted on this particular zone to elucidate the composition of newly formed layer. It was revealed that layer was primarily

consisted of Al, O, and Cr as major constituents. Additionally, presence of Ni and Cr elements confirmed their origin from the Nimonic substrate while Si, Al, and O elements were attributed to the composite bond coat. Furthermore, Zr inclusions were observed in the composite bond coat.



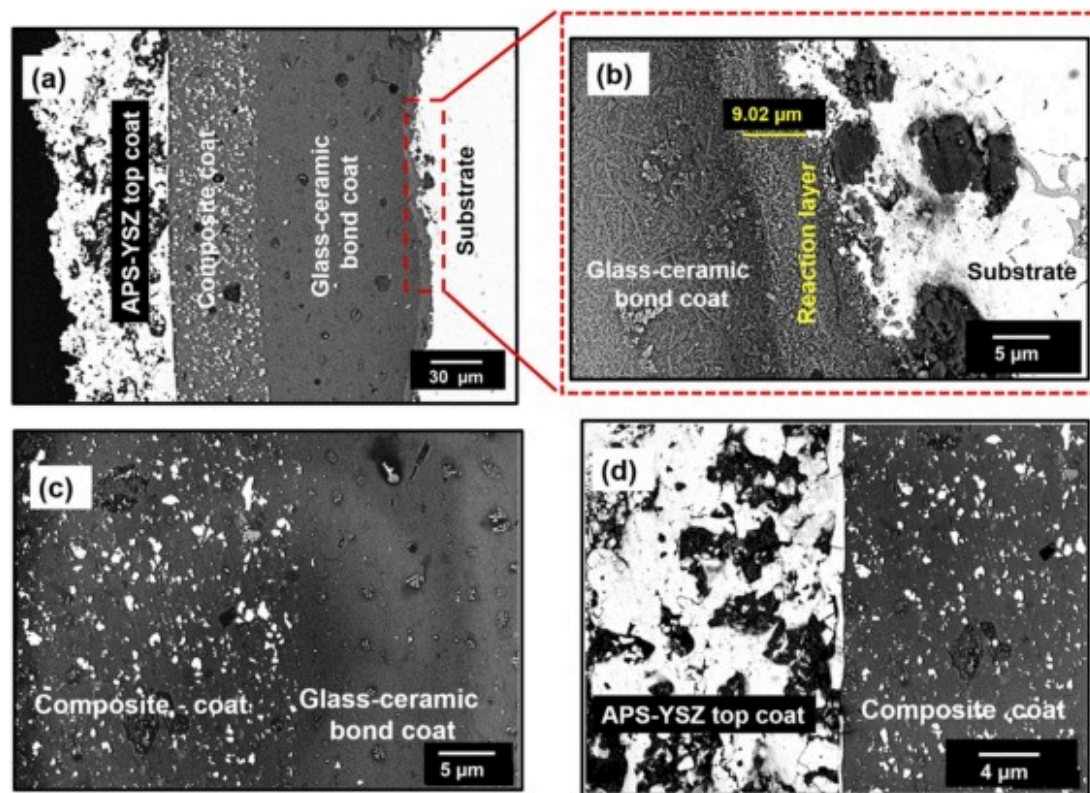
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Fig. 9. (a) Cross-sectional FESEM micrograph of composite bond coated duplex layered TBC system, (b) substrate-bond coat interface (magnified) and (c) bond coat-top coat after oxidation (magnified) after 100 h oxidation at 1000 °C.

Fig. 10 shows cross-sectional microstructure of double bond coated TBC3 system after isothermal oxidation for 100 h at 1000 °C showing all the three layers. A clear interface was evident between bond coat and composite intermediate layer (Fig. 10c), indicating a gradual transition between the layers. Consequently, no reaction was observed between them. Furthermore, similar to previous observations, there was no layer between top coat and underlying composite coat (Fig. 10d). However, a reaction layer (thickness $\approx 9 \mu\text{m}$) was observed between bond coat and substrate (Fig. 10b). Reaction layer formation suggested limited

diffusion of oxygen ions from the oxidation environment through composite coat and bond coat to substrate resulting in development of thinner reaction scale at substrate-bond coat interface. Thickness of reaction layer was $\approx 16 \mu\text{m}$ for glass-ceramic bond coated TBC system while use of composite layer as bond coat reduced the reaction layer thickness to $\approx 14 \mu\text{m}$. On the contrary, by stacking glass-ceramic coat and composite coat one above the other and forming a three layered TBC system the oxygen penetration was successfully controlled, which resulted in further decrease in the reaction layer thickness (thickness $\approx 9 \mu\text{m}$). The developed reaction layer primarily consisted of refractory scale made of alumina and chromia. The strain associated with the formation of that new layer within the TBC system could lead to internal stress-related issues during prolonged operation. Therefore, determining the optimum thickness of the reaction layer was crucial for maintaining a well-sustained TBC system. Ongoing research reduced thickness of reaction layer at bond coat-substrate interfacial region by employing glass-ceramic based double bond coated TBC system.



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Fig. 10. (a) Cross-sectional FESEM micrograph of triple layered TBC system, (b) substrate-bond coat interface (magnified), (c) bond coat-composite intermediate coat interface (magnified) and (d) composite intermediate coat-top coat interface (magnified) after 100 h oxidation at 1000 °C.

Based on long term oxidation it can be said that TBC3 could be effective as reaction layer thickness could be reduced considerably by using glass-ceramic bond coat and composite coat as double bond coats between substrate and top coat [48]. Oxide layer between substrate and bond coat caused residual stress generation. Composite coat of TBC3 system accommodated the residual stresses through creep due to relatively higher CTE value than glass-ceramic bond coat and top coat. Therefore, top coat chipping could be lowest in TBC3 system [49]. It is noteworthy that unlike conventional (NiCoCrAlY/NiCrAlY) bond coated TBC systems, glass-ceramic based single and double bond coated TBC systems exhibited complete elimination of TGO layer, which is very advantageous in terms of reliability and durability of a TBC system.

3.5. Mechanical properties

TBC3 system isothermally oxidized at 1000 °C showed that glass-ceramic bond coat, glass-ceramic/YSZ composite coat and YSZ top coat had nano hardness value of 10.0 ± 0.4 GPa, 9.2 ± 1.1 GPa, 13.6 ± 0.1 GPa, respectively. These coatings also showed the Young's modulus value of 99.9 ± 3.0 GPa, 104.1 ± 1.0 GPa, 124.2 ± 0.6 GPa, respectively. Fracture toughness of as prepared and oxidized glass-ceramic coating was measured to be 0.651 ± 0.15 MPa.m^{1/2} and 1.0005 ± 0.25 MPa.m^{1/2}, respectively. Fracture toughness of as prepared and oxidized composite coating was measured to be 0.908 ± 0.19 MPa.m^{1/2} and 0.920 ± 0.21 MPa.m^{1/2}, respectively. Fracture toughness of as prepared and oxidized top coat was measured to be 0.623 ± 0.22 MPa.m^{1/2} and 0.154 ± 0.27 MPa.m^{1/2}, respectively. Oxidized coating had higher fracture toughness due to oxidation induced stress present in the coating, which is well consistent with available data [50]. The surface roughness of these oxidized coatings was found to be 3.26 µm, 4.85 µm and 24.80 µm, respectively. These glass-ceramic and composite coatings exhibited good mechanical properties due to presence of refractory several crystalline phases like periclase, mullite, magnesium aluminate spinel and zircon phase. Additionally, these coatings demonstrate strong adherence through an interlocking microstructure, as observed in the Fig. 4 showing morphology. The surface roughness further enhances their adherence, preventing delamination even under prolonged oxidation conditions

Experimental results showed that glass-ceramic bond coatings demonstrated excellent oxidation resistance and effectively prevented development of TGO between top coat and bond coat. EDX analysis (Fig. 6, Fig. 7) confirmed the presence of protective

scales on the substrates of both duplex and triplex TBC systems. Interestingly, the analysis revealed that protective scales were more prominent in the duplex layered TBC compared to triplex TBC system. Further examination of cross-section morphologies (Fig. 6, Fig. 7) suggested that the highlighted areas predominantly consisted of Al_2O_3 , as indicated by presence of O and Al elements in the EDX results. Gradual formation of the protective scale (Al_2O_3) may explain the steady increase in mass as observed in the triplex system during oxidation for 100 h at 1000°C (Fig. 3).

TGO in traditional TBC system forms between metallic bond coat and ceramic top coat that grows significantly in an uncontrolled way during long thermal exposure resulting in enormous stress, which generates cracks and eventually leads to spallation of TBC [51]. However, similar TGO was absent at the interface of bond coat and top coat or composite coat and top coat in the new TBC systems. Rather thin oxide layer was present between metal substrate and oxide based bond coat to aid adherence as well as prevent oxidation. It was easier to control the growth of thin oxide layer formed within new TBC systems.

4. Conclusions

This paper investigates the isothermal oxidation behaviour of three distinct TBC systems. The conclusions drawn from this study are summarized as follows:

- Three glass-ceramic based TBC systems did not show any thermally grown oxide (TGO) layer at bond coat-top coat interface.
- Protective scale predominantly composed of alumina was formed between substrate and bond coat during oxidation for 100 h at 1000 °C, which significantly increased bond coat's adhesion to the substrate.
- Triple-layered TBC system demonstrated best oxidation resistance up to 1000°C due to controlled weight change compared to the other TBC systems.
- The variation in oxidation behaviour between duplex and triplex TBC systems was linked to differences in microstructure affecting oxygen ingress into the zirconia top layer and thereafter permeation through the

underlying coating. Overall, present research findings indicated critical role of coating composition and microstructure of TBC systems in determining their oxidation performance under high-temperature exposure.

CRedit authorship contribution statement

Sumana Ghosh: Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Everson Kandare:** Writing – review & editing, Supervision. **Sandip Bysakh:** Writing – review & editing, Methodology. **Karthiga Parthiban:** Writing – original draft, Methodology, Investigation, Data curation. **Abhijit Date:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Karthiga Parthiban reports financial support was provided by Council for Scientific and Industrial Research. Karthiga Parthiban reports a relationship with RMIT University that includes: NA has patent pending to NA. NA If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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